James Bowden

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EDUCATION

California Institute of Technology **2019 – Jun 2023**
B.S. Computer Science, Data Science Minor GPA: 4.0

EXPERIENCE

Ryan Adams Group (LIPS), Princeton **Jun 2022 – Present**

- Reframing topology optimization of mechanical structures with deep implicit surfaces to be able to specify functional design priors
- Creating end-to-end differentiable pipeline for rational design of 2/3-D materials given arbitrary mechanical objectives, e.g., precise multi-stability in metamaterials

Katie Bouman Group, Caltech **Jan 2022 – Present**

- Building from CycleGAN to exaggerate important decision features in image data for model explainability and design of models consistent with domain-expert priors

Anima Anandkumar Group, Caltech **Jan – Mar 2022**

- Worked on efficiently training diffusion models by partitioning ~200K-D data in Fourier domain and hierarchically conditioning high-res. signals on lower-frequency (global) signals
- Cut model sampling time from ~14hrs to ~2hrs in early tests by replacing convolutional network ("overfit" to images) with simpler MLP architectures (ReZero-net) for freq. domain

Yisong Yue Group, Caltech **Jun 2020 – Present**

- Developing (multi-fidelity) DK-BO for real-world rational design applications including COVID antibody engineering and nanophotonics filter design in collaboration with LLNL, UChicago
- Integrated Deep Kernel Learning (DKL) with BayesOpt (BO) to find global optima faster via improved model fitting on complex, high-dimensional datasets where GPs struggle
- Created "fully-Bayesian" active learning pipeline with posterior sampling methods (Thompson sampling and Monte Carlo dropout)
- Helped develop learned region-of-interest DK-BO to adaptively filter candidate datapoints

Software Engineering Intern, Uber Driver Trip Pricing **Jun – Sept 2021**

- Designed conditional models to increase driver offer acceptance, trip completion rates
- Pioneered geo-time embedding to create country-wide model with cold-start prediction capabilities for transfer learning

Head Teaching Assistant, Caltech **Sept 2020 – Present**

- Hold office hours, write assignments, give code reviews, lead lab section, grade code, instill good development practices. As Head TA for CS 2, hire/train/support/manage 18 other TAs.

  CS 156ab: **Machine Learning** [FA 21/22 SP 22/23] • CS 24: **Computing Systems** [FA 21] • CS 3: **Software Design** [SP 21] • CS 2: **Data Structures/Algorithms** [WI 21/22/23] • CS 1: **Intro CS** [FA 20]

PUBLICATIONS

Learning Region of Interest for Bayesian Optimization with Adaptive Level-Set Estimation
F. Zhang, J. Song, J. Bowden, A. Ladd, Y. Yue, T. Desautels, Y. Chen.
ICML ReALML Workshop 2022. https://realworldml.github.io/files/cr/paper63.pdf

Deep Kernel Bayesian Optimization
J. Bowden, J. Song, Y. Chen, Y. Yue, T. Desautels.
Pre-print 2021. https://www.osti.gov/biblio/1811769-deep-kernel-bayesian-optimization

Bridge-Group: An Opt-In Recitation Section to Facilitate the Transition from CS1 to CS2
E. Gurcan, J. Bowden, A. Blank.
RESPECT 2022. https://james-bowden.github.io/pages/teaching/bridge_group

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EXPERIENCE, CONT'D

ML Researcher, Caltech (Independent Group Project) **Mar – Jun 2022**

- Developed domain-aware data augmentation techniques for graph-structured data to better train GNNs, particularly for molecule data predictive of HIV binding effectivity
- Generated *de novo* graphs to make model invariant to inner nodes (less influence on binding) by randomly replacing central nodes with similar atoms, and invariant to non-binding outer nodes by randomly adding small atoms to degree 3 nodes with some probability
- Selected subgraphs to ignore/collapse interior chemical rings less likely to influence binding

ML Researcher, Caltech (Independent Group Project) **Mar – Jun 2021**

- Devised NAS method using PAC-Bayes theory that utilizes untrained architectures to evaluate objectives 50x faster with randomly sampled weights as performance proxy
- Designed perturbation objective employing Monte Carlo dropout to approximate α-robustness and local weight space flatness; this objective was predictive of test accuracy

ML Engineer, Caltech (Group Projects) **Jan – Mar 2021**

- Wildfire prediction: analyzed and preprocessed noisy data, tested/tuned a variety of model types with limited queries to the test dataset. Chose soft voting ensembles to reduce overall variance while still using strong models with low bias.
- Movie rating dataset: performed analysis and unsupervised learning (matrix factorization) to generate insights and clustering for movie dataset. Found that genres and popular actors were often mapped to similar regions in our 2D representation.
- Sonnet generation: tokenized words in training set of Shakespearean sonnets and generated look-alike poems with Hidden Markov Models, RNNs (LSTM) that rhymed!

Kaihang Wang Lab, Caltech **Dec 2019 – Jan 2021**

- Led team of 3 senior students to create automated data pipeline for reconstructing ancestral genomes using Python and integrating various command line tools
- Wrote Python tools to inform and assist wet bench projects including recoding genomes and predicting minimum essential genes for life

Software Developer, Caltech (Group Project) **Mar – Jun 2020**

- Created escape game in C, from scratch, with A* pathfinding, physics engine, vision, map
- Significantly reduced lag via dynamic programming

Pamela Bjorkman Lab, Caltech **Aug – Dec 2019**

- Developed multivalent nanoparticles with a variety of conserved flu/HIV antigenic regions to elicit bnAbs, neutralizing antibodies effective across different virus strains

Jerry Shay Lab, UT Southwestern Medical Center **May – Aug 2019**

- Showed that NIP45 is first known protein in ALT cancer repertoire to bind SUMO1, implying involvement in formation of PML bodies for telomere extension

AWARDS

Rypisi SURF Fellow, 2022. Awarded by the director of Caltech's summer research program.

Associates SURF Fellow, 2020. For outstanding research.

Teaching Mode, 2018. For best research presentation in cohort of 30 students.

Thermo-Fisher Scholarship, 2019. For biomedical research (1 of 6).